MATHEMATICS BASIC 2023 ALL SETS

Chapter 1: Real Numbers

YEAR: 2023
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The prime factorisation of natural number 288 is
OPTIONS: (A) $2^4 \times 3^3$ (B) $2^4 \times 3^2$ (C) $2^5 \times 3^2$ (D) $2^5 \times 3^1$
ANSWER: (C) $2^5 \times 3^2$

YEAR: 2023
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The prime factorisation of 1728 is
OPTIONS: (A) $2^5 \times 3^3$ (B) $2^5 \times 3^4$ (C) $2^6 \times 3^3$ (D) $2^6 \times 3^2$
ANSWER: (C) $2^6 \times 3^3$

YEAR: 2023
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The prime factorisation of 432 is
OPTIONS: (A) $2^3 \times 3^4$ (B) $2^4 \times 3^3$ (C) $2^3 \times 3^3$ (D) $2^4 \times 3^4$
ANSWER: (B) $2^4 \times 3^3$

YEAR: 2023
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Find the LCM and HCF of 92 and 510, using prime factorisation.
ANSWER: $92 = 2^2 \times 23$, $510 = 2 \times 3 \times 5 \times 17$. HCF = 2, LCM = 23460

YEAR: 2023
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Find the HCF of the numbers 540 and 630, using prime factorization method.
ANSWER: $540 = 2^2 \times 3^3 \times 5$, $630 = 2 \times 3^2 \times 5 \times 7$. HCF $= 2 \times 3^2 \times 5 = 90$

YEAR: 2023
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Show that $(15)^n$ cannot end with the digit 0 for any natural number n.
ANSWER: $15^n = 3^n \times 5^n$. For a number to end with zero it should have both 2 and 5 in its prime factorization. 15^n has only 3 and 5 as factors, so it cannot end with zero.

YEAR: 2023
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Prove that $3 + 7\sqrt{2}$ is an irrational number, given that $\sqrt{2}$ is an irrational number.
ANSWER: Let $3 + 7\sqrt{2} = p/q$ where p and q are integers, $q \neq 0$. Then $\sqrt{2} = (p-3q)/7q$. RHS is rational but LHS is irrational. Hence contradiction. So $3 + 7\sqrt{2}$ is irrational.

YEAR: 2023
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Prove that $5 - \sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is an irrational number.
ANSWER: Let $5 - \sqrt{3}$ be rational. Then $\sqrt{3} = 5$ - rational, so $\sqrt{3}$ would be rational, which is a contradiction. Hence $5 - \sqrt{3}$ is irrational.

YEAR: 2023
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3

QUESTION: Prove that $4 + 2\sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is an irrational number.
ANSWER: Let $4 + 2\sqrt{3} = p/q$. Then $\sqrt{3} = (p - 4q)/2q$. RHS is rational, but LHS is irrational. Contradiction. Hence $4 + 2\sqrt{3}$ is irrational.

YEAR: 2023
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Prove that $2 - 3\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is an irrational number.
ANSWER: Let $2 - 3\sqrt{5} = p/q$. Then $\sqrt{5} = (2q - p)/3q$. RHS is rational, but LHS is irrational. Contradiction. Hence $2 - 3\sqrt{5}$ is irrational.

YEAR: 2023
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Prove that $3 + 7\sqrt{2}$ is an irrational number, given that $\sqrt{2}$ is an irrational number.
ANSWER: Let $3 + 7\sqrt{2}$ be rational. Then $7\sqrt{2}$ = rational - 3, so $\sqrt{2}$ would be rational. Contradiction. Hence $3 + 7\sqrt{2}$ is irrational.

YEAR: 2023
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The prime factorisation of the number 5488 is
OPTIONS: (A) $2^3 \times 7^3$ (B) $2^4 \times 7^3$ (C) $2^4 \times 7^4$ (D) $2^3 \times 7^4$
ANSWER: (B) $2^4 \times 7^3$

YEAR: 2023
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The prime factorisation of the number 2304 is
OPTIONS: (A) $2^8 \times 3^2$ (B) $2^7 \times 3^3$ (C) $2^8 \times 3^1$ (D) $2^7 \times 3^2$
ANSWER: (A) $2^8 \times 3^2$

YEAR: 2023
CHAPTER: 1 - Real Numbers
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: If n is a natural number, then 8^n cannot end with digit
OPTIONS: (A) 0 (B) 2 (C) 4 (D) 6
ANSWER: (A) 0

Chapter 2: Polynomials

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The zeroes of the polynomial $p(x) = x^2 + 3x + 2$ are given as.
OPTIONS: (A) 1,2 (B) 2,-1 (C) -2,1 (D) -2,-1
ANSWER: (D) -2, -1

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The sum and product of zeroes of the polynomial $p(x) = 3x^2 - 5x + 2$ are
OPTIONS: (A) 5/3, 2/3 (B) -5/3, 2/3 (C) 1, 2/3 (D) -5/3, -2/3
ANSWER: (A) 5/3, 2/3

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The graph of $y = p(x)$ is shown in the figure for some polynomial $p(x)$. The number of zeroes of $p(x)$ is/are:
OPTIONS: (A) 0 (B) 1 (C) 2 (D) 3
ANSWER: (A) 0

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Graph of a polynomial p(x) is given in the figure. The number of zeroes of p(x) is:
OPTIONS: (A) 2 (B) 3 (C) 4 (D) 5
ANSWER: (A) 2

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The zeroes of the polynomial p(x) = 2x^2 - x - 3 are
OPTIONS: (A) -3/2, 1 (B) 3/2, 1 (C) -3/2, -1 (D) 3/2, -1
ANSWER: (D) 3/2, -1

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: If (-3) is one of the zeroes of the polynomial (k-1)x^2 + kx + 1, find the value of k.
ANSWER: (k-1)(-3)^2 + k(-3) + 1 = 0 => 9k - 9 - 3k + 1 = 0 => 6k = 8 => k = 4/3

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Find the value of k such that the polynomial p(x) = 3x^2 + 2kx + x - k - 5 has the sum of zeroes equal to half of their product.
ANSWER: 3x^2 + (2k+1)x - k - 5. Sum = -(2k+1)/3, Product = -(k+5)/3. Sum = 1/2 × Product => -(2k+1)/3 = -(k+5)/6 => 2(2k+1) = k+5 => 4k+2 = k+5 => 3k = 3 => k = 1

YEAR: 2023

CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: If α, β are zeroes of the quadratic polynomial x^2 + 3x + 2, find a quadratic polynomial whose zeroes are α+1, β+1.
ANSWER: α+β = -3, αβ = 2. Sum of new zeroes = (α+1)+(β+1) = -3+2 = -1. Product = (α+1)(β+1) = αβ+α+β+1 = 2-3+1 = 0. Required polynomial = x^2 + x.

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Find the zeroes of the polynomial p(x) = 2x^2 - 7x - 15 and verify the relationship between its coefficients and zeroes.
ANSWER: 2x^2 - 7x - 15 = (2x+3)(x-5). Zeroes are -3/2 and 5. Sum = 7/2 = -coefficient of x/coefficient of x^2. Product = -15/2 = constant term/coefficient of x^2.

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Find the zeroes of the polynomial p(x) = 3x^2 + 5x - 28 and verify the relationship between its coefficients and zeroes.
ANSWER: 3x^2 + 5x - 28 = (3x-7)(x+4). Zeroes are 7/3 and -4. Sum = 7/3 - 4 = -5/3 = -coefficient of x/coefficient of x^2. Product = 7/3 × (-4) = -28/3 = constant term/coefficient of x^2.

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: If α, β are zeroes of the quadratic polynomial x^2 - 5x + 6, form another quadratic polynomial whose zeroes are 1/α and 1/β.
ANSWER: α+β = 5, αβ = 6. Sum = 1/α + 1/β = (α+β)/αβ = 5/6. Product = 1/(αβ) = 1/6. Required polynomial = x^2 - 5/6x + 1/6 or 6x^2 - 5x + 1.

YEAR: 2023

CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Find the zeroes of the polynomial $p(x) = 3x^2 + 5x - 28$ and verify the relationship between its coefficients and zeroes.
ANSWER: $p(x) = 3x^2 + 5x - 28 = (3x-7)(x+4)$. Zeroes are 7/3 and -4. Sum = 7/3 - 4 = -5/3 = -coefficient of x/coefficient of x^2. Product = 7/3 × (-4) = -28/3 = constant term/coefficient of x^2.

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: If one zero of the quadratic polynomial $kx^2 + 3x + k$ is 2, then the value of k is:
OPTIONS: (A) -6/5 (B) 6/5 (C) 5/6 (D) -5/6
ANSWER: (A) -6/5

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The zeroes of the quadratic polynomial $16x^2 - 9$ are:
OPTIONS: (A) 3/4, 3/4 (B) -3/4, 3/4 (C) 9/16, 9/16 (D) -3/4, -3/4
ANSWER: (B) -3/4, 3/4

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A quadratic polynomial the sum and product of whose zeroes are -3 and 2 respectively, is:
OPTIONS: (A) $x^2 + 3x + 2$ (B) $x^2 - 3x + 2$ (C) $x^2 - 3x - 2$ (D) $x^2 + 3x - 2$
ANSWER: (A) $x^2 + 3x + 2$

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1
QUESTION: A quadratic polynomial whose sum and product of zeroes are 2 and -1 respectively is:
OPTIONS: (A) $x^2 + 2x + 1$ (B) $x^2 - 2x - 1$ (C) $x^2 + 2x - 1$ (D) $x^2 - 2x + 1$
ANSWER: (B) $x^2 - 2x - 1$

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: α, β are the zeroes of the quadratic polynomial $p(x) = x^2 - 8x + k$, such that $\alpha^2 + \beta^2 = 40$. Find the value of k.
ANSWER: $\alpha+\beta = 8$, $\alpha\beta = k$. $\alpha^2+\beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 64 - 2k = 40 \Rightarrow 2k = 24 \Rightarrow k = 12$.

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The zeroes of the polynomial $p(x) = 25x^2 - 49$ are:
OPTIONS: (A) 49/25, 49/25 (B) -49/25, +49/25 (C) 7/5, -7/5 (D) 7/5, 7/5
ANSWER: (C) 7/5, -7/5

Chapter 3: Pair of Linear Equations in Two Variables

YEAR: 2023
CHAPTER: 3 - Pair of Linear Equations in Two Variables
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The pair of linear equations $3x + 5y - 4 = 0$ and $15x + 25y - 25 = 0$ is inconsistent.
OPTIONS: (A) Assertion true, Reason true and correct explanation (B) Both true but Reason not correct explanation (C) Assertion true, Reason false (D) Assertion false, Reason true
ANSWER: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation

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YEAR: 2023
CHAPTER: 3 - Pair of Linear Equations in Two Variables
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Solve for x and y: $x + y = 6$, $2x - 3y = 4$.
ANSWER: Solving: $x = 22/5$, $y = 8/5$

YEAR: 2023
CHAPTER: 3 - Pair of Linear Equations in Two Variables
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Find out whether the following pair of linear equations are consistent or inconsistent: $5x - 3y = 11$, $-10x + 6y = 22$
ANSWER: $a1/a2 = 5/-10 = -1/2$, $b1/b2 = -3/6 = -1/2$, $c1/c2 = 11/22 = 1/2$. Since $a1/a2 = b1/b2 \neq c1/c2$, the pair is inconsistent.

YEAR: 2023
CHAPTER: 3 - Pair of Linear Equations in Two Variables
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: A fraction becomes 1 when 1 is added to the numerator and 1 is subtracted from the denominator. If we add 1 to the denominator, it becomes 1/2. Find the fraction.
ANSWER: Let fraction be x/y. $(x+1)/(y-1) = 1 \Rightarrow x+1 = y-1 \Rightarrow x - y = -2$. $x/(y+1) = 1/2 \Rightarrow 2x = y+1 \Rightarrow 2x - y = 1$. Solving: $x = 3$, $y = 5$. Fraction = 3/5.

YEAR: 2023
CHAPTER: 3 - Pair of Linear Equations in Two Variables
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: For which value of k will the following pair of linear equations have no solution? $3x + y = 1$, $(2k-1)x + (k-1)y = 2k+1$
ANSWER: For no solution: $a1/a2 = b1/b2 \neq c1/c2$. $3/(2k-1) = 1/(k-1) \Rightarrow 3k-3 = 2k-1 \Rightarrow k = 2$. Check: $3/(3) = 1/(1) = 1 \neq c1/c2 = 1/5$. So $k = 2$.

YEAR: 2023
CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The pair of linear equations $x + 2y - 5 = 0$ and $2x - 4y + 6 = 0$:
OPTIONS: (A) is inconsistent (B) is consistent with many solutions (C) is consistent with a unique solution (D) is consistent with two solutions
ANSWER: (C) is consistent with a unique solution

YEAR: 2023
CHAPTER: 3 - Pair of Linear Equations in Two Variables
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The lines represented by the linear equations $y = x$ and $x = 4$ intersect at P. The coordinates of the point P are:
OPTIONS: (A) (4,0) (B) (4,4) (C) (0,4) (D) (-4,4)
ANSWER: (B) (4,4)

YEAR: 2023
CHAPTER: 3 - Pair of Linear Equations in Two Variables
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The pair of linear equations $x + 2y + 5 = 0$ and $-3x - 6y + 1 = 0$ has:
OPTIONS: (A) a unique solution (B) exactly two solutions (C) infinitely many solutions (D) no solution
ANSWER: (D) no solution

YEAR: 2023
CHAPTER: 3 - Pair of Linear Equations in Two Variables
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The value of k for which the equations $3x - y + 8 = 0$ and $6x - ky + 16 = 0$ represent coincident lines is:
OPTIONS: (A) 1/2 (B) -1/2 (C) 2 (D) -2
ANSWER: (C) 2

YEAR: 2023
CHAPTER: 3 - Pair of Linear Equations in Two Variables

ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Find the values of a and b for which the system of linear equations $3x + 4y = 12$, $(a+b)x + 2(a-b)y = 24$ has infinite number of solutions.
ANSWER: For infinite solutions: $3/(a+b) = 4/[2(a-b)] = 12/24 = 1/2$. So $a+b = 6$, $a-b = 4$. Solving: $a = 5$, $b = 1$.

YEAR: 2023
CHAPTER: 3 - Pair of Linear Equations in Two Variables
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: A lending library has a fixed charge for first three days and an additional charge for each day thereafter. Rittik paid ₹27 for a book kept for 7 days and Manmohan paid ₹21 for a book kept for 5 days. Find the fixed charges and the charge for each extra day.
ANSWER: Let fixed charge = x, charge per extra day = y. $x + 4y = 27$, $x + 2y = 21$. Solving: $2y = 6 \Rightarrow y = 3$, $x = 15$. Fixed charge = ₹15, charge per extra day = ₹3.

YEAR: 2023
CHAPTER: 3 - Pair of Linear Equations in Two Variables
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Find the values of a and b for which the system of linear equations $3x + 4y = 12$, $(a+b)x + 2(a-b)y = 24$ has infinite number of solutions.
ANSWER: For infinite solutions: $3/(a+b) = 4/[2(a-b)] = 12/24 = 1/2$. So $a+b = 6$, $a-b = 4$. Solving: $a = 5$, $b = 1$.

Chapter 4: Quadratic Equations

YEAR: 2023
CHAPTER: 4 - Quadratic Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The discriminant of the quadratic equation $2x^2 - 5x - 3 = 0$ is
OPTIONS: (A) 1 (B) 49 (C) 7 (D) 19
ANSWER: (B) 49

YEAR: 2023
CHAPTER: 4 - Quadratic Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The discriminant of the quadratic equation $2x^2 + x - 1 = 0$ is:
OPTIONS: (A) -9 (B) -7 (C) 9 (D) 7
ANSWER: (C) 9

YEAR: 2023
CHAPTER: 4 - Quadratic Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: If the quadratic equation $9x^2 + bx + 1/4 = 0$ has equal roots, then the value of b is:
OPTIONS: (A) 0 (B) -3 only (C) 3 only (D) ± 3
ANSWER: (D) ± 3

YEAR: 2023
CHAPTER: 4 - Quadratic Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A quadratic equation whose one root is 2 and the sum of whose roots is zero, is:
OPTIONS: (A) $x^2 + 4 = 0$ (B) $x^2 - 2 = 0$ (C) $4x^2 - 1 = 0$ (D) $x^2 - 4 = 0$
ANSWER: (D) $x^2 - 4 = 0$

YEAR: 2023
CHAPTER: 4 - Quadratic Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following is not a quadratic equation?
OPTIONS: (A) $2(x-1)^2 = 4x^2 - 2x + 1$ (B) $2x - x^2 = x^2 + 5$ (C) $(\sqrt{2}x + \sqrt{3})^2 + x^2 = 3x^2 - 5x$ (D) $(x^2+2x)^2 = x^4 + 3 + 4x^3$
ANSWER: (C) $(\sqrt{2}x + \sqrt{3})^2 + x^2 = 3x^2 - 5x$

YEAR: 2023

CHAPTER: 4 - Quadratic Equations
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: The sum of reciprocals of Roohi's age (in years) 3 years ago and 5 years hence from now is 1/3. Find her present age.
ANSWER: Let present age = x. 1/(x-3) + 1/(x+5) = 1/3 => (2x+2)/(x^2+2x-15) = 1/3 => 6x+6 = x^2+2x-15 => x^2 - 4x - 21 = 0 => (x-7)(x+3) = 0 => x = 7.

YEAR: 2023
CHAPTER: 4 - Quadratic Equations
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: A train travels 360 km at a uniform speed. If the speed had been 5 km/hr more, it would have taken 1 hour less for the same journey. Find the speed of the train.
ANSWER: Let usual speed = x km/hr. 360/x - 360/(x+5) = 1 => 360(x+5-x)/[x(x+5)] = 1 => 1800 = x^2+5x => x^2+5x-1800 = 0 => (x+45)(x-40) = 0 => x = 40 km/hr.

YEAR: 2023
CHAPTER: 4 - Quadratic Equations
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Find the value of k for which the roots of the quadratic equation 5x^2 - 10x + k = 0 are real and equal.
ANSWER: For equal roots: D = 0 => (-10)^2 - 4(5)(k) = 0 => 100 - 20k = 0 => k = 5

YEAR: 2023
CHAPTER: 4 - Quadratic Equations
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: If one root of the quadratic equation 3x^2 - 8x - (2k+1) = 0 is seven times the other, find the value of k.
ANSWER: Let roots be α and 7α. α + 7α = 8/3 => 8α = 8/3 => α = 1/3. α(7α) = -(2k+1)/3 => 7α^2 = -(2k+1)/3 => 7/9 = -(2k+1)/3 => 7/3 = -(2k+1) => 2k = -10/3 => k = -5/3

YEAR: 2023
CHAPTER: 4 - Quadratic Equations

ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Find the discriminant of the quadratic equation 3x^2 - 2x + 1/3 = 0 and hence find the nature of its roots.
ANSWER: D = (-2)^2 - 4(3)(1/3) = 4 - 4 = 0. Therefore roots are real and equal.

YEAR: 2023
CHAPTER: 4 - Quadratic Equations
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Find the roots of the quadratic equation x^2 - x - 2 = 0.
ANSWER: x^2 - x - 2 = (x-2)(x+1). Roots are 2 and -1.

YEAR: 2023
CHAPTER: 4 - Quadratic Equations
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: The sum of the ages of a father and his son is 45 years. Five years ago, the product of their ages (in years) was 124. Determine their present age.
ANSWER: Let father's age = x, son's age = 45-x. (x-5)(40-x) = 124 => -x^2+45x-324 = 0 => x^2-45x+324 = 0 => (x-36)(x-9) = 0 => x = 36 (reject 9). Father = 36 years, Son = 9 years.

Chapter 5: Arithmetic Progressions

YEAR: 2023
CHAPTER: 5 - Arithmetic Progressions
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The seventh term of an A.P. whose first term is 28 and common difference -4, is
OPTIONS: (A) 0 (B) 4 (C) 52 (D) 56
ANSWER: (B) 4

YEAR: 2023
CHAPTER: 5 - Arithmetic Progressions

ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In an AP, if d = -4, n = 7 and a_n = 4, then the value of a is
OPTIONS: (A) 6 (B) 7 (C) 20 (D) 28
ANSWER: (D) 28

YEAR: 2023
CHAPTER: 5 - Arithmetic Progressions
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: How many terms are there in the A.P. given below? 14, 19, 24, 29, ..., 119
OPTIONS: (A) 18 (B) 14 (C) 22 (D) 21
ANSWER: (C) 22

YEAR: 2023
CHAPTER: 5 - Arithmetic Progressions
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The 20th term of an A.P, whose first term is -2 and the common difference is 4, is
OPTIONS: (A) 78 (B) 74 (C) -36 (D) -34
ANSWER: (B) 74

YEAR: 2023
CHAPTER: 5 - Arithmetic Progressions
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: If -5, x, 3 are three consecutive terms of an A.P., then the value of x is
OPTIONS: (A) -2 (B) 2 (C) 1 (D) -1
ANSWER: (D) -1

YEAR: 2023
CHAPTER: 5 - Arithmetic Progressions
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: If the sum of the first 7 terms of an A.P. is -14 and that of 11 terms is -55, then find the sum of its first n terms.

ANSWER: S7 = 7/2(2a+6d) = -14 => 2a+6d = -4 => a+3d = -2. S11 = 11/2(2a+10d) = -55 => 2a+10d = -10 => a+5d = -5. Solving: 2d = -3 => d = -3/2, a = 5/2. Sn = n/2[2(5/2)+(n-1)(-3/2)] = n/2[5 - 3n/2 + 3/2] = n/4[13 - 3n]

YEAR: 2023
CHAPTER: 5 - Arithmetic Progressions
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: In an A.P., the sum of the first n terms is 3n^2 + n. Find the first term and the common difference of the A.P. Hence, find its 15th term.
ANSWER: S_n = 3n^2 + n. a = S1 = 3+1 = 4. S2 = 12+2 = 14. a2 = S2 - S1 = 14-4 = 10. d = 6. a15 = a + 14d = 4 + 84 = 88.

YEAR: 2023
CHAPTER: 5 - Arithmetic Progressions
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: The sum of the first 7 terms of an A.P. is -21 and that of the first 17 terms is -221, find the sum of its first n terms.
ANSWER: S7 = 7/2(2a+6d) = -21 => 2a+6d = -6 => a+3d = -3. S17 = 17/2(2a+16d) = -221 => 2a+16d = -26 => a+8d = -13. Solving: 5d = -10 => d = -2, a = 3. Sn = n/2[6+(n-1)(-2)] = n/2[6-2n+2] = n/2[8-2n] = n(4-n)

YEAR: 2023
CHAPTER: 5 - Arithmetic Progressions
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: A man repays a loan of ₹3,250 by paying ₹20 in the first month and then increases the payment by ₹15 every month. How long will it take to clear the loan?
ANSWER: a = 20, d = 15, S_n = 3250. n/2[40+(n-1)15] = 3250 => n(40+15n-15) = 6500 => 15n^2+25n-6500 = 0 => 3n^2+5n-1300 = 0 => (3n+65)(n-20) = 0 => n = 20 months.

YEAR: 2023
CHAPTER: 5 - Arithmetic Progressions
ORIGINAL_SECTION: E
ORIGINAL_MARKS: 4

QUESTION: Aahana placed pots in such a way that number of pots in the first row is 2, second row is 5, third row is 8 and so on. (i) Find the number of pots placed in the 10th row. (ii) Find the difference in the number of pots placed in 5th row and 2nd row. (iii) If Aahana wants to place 100 pots in total, find the total number of rows formed in the arrangement. OR (iii) If Aahana has sufficient space for 12 rows, how many total pots are placed?
ANSWER: a = 2, d = 3. (i) a10 = 2+9(3) = 29. (ii) a5 - a2 = (2+4×3) - (2+1×3) = 14-5 = 9. (iii) Sn = n/2[4+(n-1)3] = 100 => n(3n+1) = 200 => 3n^2+n-200 = 0 => (3n+25)(n-8) = 0 => n = 8. OR S12 = 12/2[4+33] = 222.

YEAR: 2023
CHAPTER: 5 - Arithmetic Progressions
ORIGINAL_SECTION: E
ORIGINAL_MARKS: 4
QUESTION: Sumant's mother started a new shoe shop. To display the shoes, she put 3 pairs of shoes in the 1st row, 5 pairs in the 2nd row, 7 pairs in the 3rd row and so on. (i) How many pairs of shoes are displayed in the 6th row? (ii) What is the difference of pairs of shoes in the 1st row and the 6th row? (iii)(a) Find the total number of pairs of shoes displayed in the first 15 rows. OR (b) If the pairs of shoes displayed in the 4th row are on sale at price of ₹500 for each pair, find the total amount earned if all shoes displayed in the 4th row are sold out.
ANSWER: a = 3, d = 2. (i) a6 = 3+5×2 = 13. (ii) Difference = 13-3 = 10. (iii)(a) S15 = 15/2[6+14×2] = 15/2[34] = 255. (b) a4 = 3+3×2 = 9. Amount = 9×500 = ₹4500.

Chapter 6: Coordinate Geometry

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The distance between the points (3, 0) and (0, -3) is
OPTIONS: (A) 2√3 units (B) 6 units (C) 3 units (D) 3√2 units
ANSWER: (D) 3√2 units

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1

QUESTION: The distance between the points (-5/2, 7) and (-1/2, 7) is:
OPTIONS: (A) 3 (B) 2 (C) 4 (D) 9
ANSWER: (B) 2

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Find the coordinates of the point which divides the line segment joining the points (7, -1) and (-3, 4) internally in the ratio 2:3.
ANSWER: P = [2(-3)+3(7)]/(2+3) = (-6+21)/5 = 3. y = [2(4)+3(-1)]/5 = (8-3)/5 = 1. Coordinates = (3,1)

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Find the value(s) of y for which the distance between the points A(3, -1) and B(11, y) is 10 units.
ANSWER: AB^2 = 100 => (11-3)^2 + (y+1)^2 = 100 => 64 + (y+1)^2 = 100 => (y+1)^2 = 36 => y+1 = ±6 => y = 5 or -7

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Show that the points A(1, 7), B(4, 2), C(-1, -1) and D(-4, 4) are vertices of the square ABCD.
ANSWER: AB = √[(4-1)^2+(2-7)^2] = √(9+25) = √34. BC = √[(-1-4)^2+(-1-2)^2] = √(25+9) = √34. CD = √[(-4+1)^2+(4+1)^2] = √(9+25) = √34. DA = √[(1+4)^2+(7-4)^2] = √(25+9) = √34. AC = √[(-1-1)^2+(-1-7)^2] = √(4+64) = √68. BD = √[(-4-4)^2+(4-2)^2] = √(64+4) = √68. All sides equal and diagonals equal, hence square.

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3
QUESTION: Prove that the points A(-1, 0), B(3, 1), C(2, 2) and D(-2, 1) are the vertices of a parallelogram ABCD. Is it also a rectangle?
ANSWER: Midpoint of AC = (1/2, 1). Midpoint of BD = (1/2, 1). Since diagonals bisect each other, ABCD is a parallelogram. AC = $\sqrt{(9+4)}$ = $\sqrt{13}$. BD = $\sqrt{(25+0)}$ = 5. Since AC ≠ BD, it is not a rectangle.

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Show that the points A(-3, 2), B(-5, -5), C(2, -3) and D(4, 4) are vertices of a rhombus ABCD. Is it also a square?
ANSWER: AB = $\sqrt{(4+49)}$ = $\sqrt{53}$. BC = $\sqrt{(49+4)}$ = $\sqrt{53}$. CD = $\sqrt{(4+49)}$ = $\sqrt{53}$. AD = $\sqrt{(49+4)}$ = $\sqrt{53}$. All sides equal, so rhombus. AC = $\sqrt{(25+25)}$ = $5\sqrt{2}$. BD = $\sqrt{(81+81)}$ = $9\sqrt{2}$. Diagonals not equal, so not a square.

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Find the co-ordinates of the points of trisection of the line-segment joining the points (5, 3) and (4, 5).
ANSWER: Let P and Q be trisection points. P divides in ratio 1:2: P = (1×4+2×5)/(3), (1×5+2×3)/(3) = (14/3, 11/3). Q divides in ratio 2:1: Q = (2×4+1×5)/(3), (2×5+1×3)/(3) = (13/3, 13/3).

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Find the value(s) of x so that PQ = QR, where the coordinates of P, Q and R are (6, -1), (1, 3) and (x, 8) respectively.
ANSWER: PQ = $\sqrt{[(1-6)^2+(3+1)^2]}$ = $\sqrt{(25+16)}$ = $\sqrt{41}$. QR = $\sqrt{[(x-1)^2+(8-3)^2]}$ = $\sqrt{[(x-1)^2+25]}$. Set PQ = QR: $(x-1)^2+25 = 41 \Rightarrow (x-1)^2 = 16 \Rightarrow x = 5$ or -3.

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: The vertices of a triangle are (-2, 0), (2, 3) and (1, -3). Is the triangle equilateral, isosceles or scalene?
ANSWER: Let A(-2,0), B(2,3), C(1,-3). AB = $\sqrt{(16+9)}$ = 5. BC = $\sqrt{(1+36)}$ = $\sqrt{37}$. CA = $\sqrt{(9+9)}$ = $3\sqrt{2}$. All sides different, so scalene.

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The distance of the point (5, 0) from the origin is
OPTIONS: (A) 0 (B) 5 (C) $\sqrt{5}$ (D) 5^2
ANSWER: (B) 5

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: If (2, 4) is the mid-point of the line-segment joining (6, 3) and (a, 5), then the value of a is
OPTIONS: (A) 2 (B) 4 (C) -4 (D) -2
ANSWER: (D) -2

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In what ratio does x-axis divide the line segment joining the points A(2, -3) and B(5, 6)?
OPTIONS: (A) 2:3 (B) 2:1 (C) 3:4 (D) 1:2
ANSWER: (D) 1:2

YEAR: 2023

CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: y-axis divides the line segment joining the points (-6, 2) and (2, -6) in the ratio:
OPTIONS: (A) 1:3 (B) 3:2 (C) 3:1 (D) 2:3
ANSWER: (C) 3:1

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The distance between the points A(0, 6) and B(-6, 2) is:
OPTIONS: (A) 6 units (B) $2\sqrt{6}$ units (C) $2\sqrt{13}$ units (D) $13\sqrt{2}$ units
ANSWER: (C) $2\sqrt{13}$ units

YEAR: 2023
CHAPTER: 6 - Coordinate Geometry
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: If the point Q(0, 1) is equidistant from the points P(5, -3) and R(x, 6), find the value of x. Also, find the distance PR.
ANSWER: PQ = QR => $(5-0)^2 + (-3-1)^2 = (x-0)^2 + (6-1)^2$ => $25+16 = x^2+25$ => $x^2 = 16$ => $x = \pm 4$. For R(4,6): PR = $\sqrt{(1^2+(-9)^2)} = \sqrt{82}$. For R(-4,6): PR = $\sqrt{(9^2+(-9)^2)} = \sqrt{162} = 9\sqrt{2}$.

Chapter 7: Triangles

YEAR: 2023
CHAPTER: 7 - Triangles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The sides of two similar triangles are in the ratio 4:7. The ratio of their perimeters is
OPTIONS: (A) 4:7 (B) 12:21 (C) 16:49 (D) 7:4
ANSWER: (A) 4:7

YEAR: 2023
CHAPTER: 7 - Triangles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In the given figure, AB || CD. If AB = 5 cm, CD = 2 cm and OB = 3 cm, then the length of OC is
OPTIONS: (A) 15/2 cm (B) 10/3 cm (C) 6/5 cm (D) 3/5 cm
ANSWER: (C) 6/5 cm

YEAR: 2023
CHAPTER: 7 - Triangles
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: In the given figure, ABC and AMP are two right triangles, right angled at B and M, respectively. Prove that ΔABC ~ ΔAMP.
ANSWER: In ΔABC and ΔAMP: ∠ABC = ∠AMP = 90°. ∠BAC = ∠MAP (common). By AA similarity, ΔABC ~ ΔAMP.

YEAR: 2023
CHAPTER: 7 - Triangles
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: S and T are points on sides PR and QR of ΔPQR such that ∠P = ∠RTS. Show that ΔRPQ ~ ΔRTS.
ANSWER: In ΔRPQ and ΔRTS: ∠R = ∠R (common). ∠P = ∠RTS (given). By AA similarity, ΔRPQ ~ ΔRTS.

YEAR: 2023
CHAPTER: 7 - Triangles
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: The diagonal BD of a quadrilateral ABCD bisects both ∠B and ∠D. Show that AB/BC = AD/CD.
ANSWER: In ΔABD and ΔCBD: ∠ABD = ∠CBD (given). ∠ADB = ∠CDB (given). By AA similarity, ΔABD ~ ΔCBD. Hence AB/BC = AD/CD.

YEAR: 2023
CHAPTER: 7 - Triangles
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: In the given figure, DE || AC and BE/EC = BC/CP. Prove that DC || AP.
ANSWER: In ΔABC, DE || AC => BE/EC = BD/DA. Given BE/EC = BC/CP => BD/DA = BC/CP. By converse of BPT, DC || AP.

YEAR: 2023
CHAPTER: 7 - Triangles
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: In the given figure, DE || AC and DF || AE. Prove that BF/FE = BE/EC.
ANSWER: In ΔABE, DF || AE => BD/DA = BF/FE (BPT). In ΔABC, DE || AC => BD/DA = BE/EC (BPT). Hence BF/FE = BE/EC.

YEAR: 2023
CHAPTER: 7 - Triangles
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: The diagonals of a quadrilateral ABCD intersect each other at the point O such that AO/BO = CO/OD. Show that quadrilateral ABCD is a trapezium.
ANSWER: AO/BO = CO/OD => OA/OC = OB/OD. ∠AOB = ∠COD (vertically opposite). By SAS similarity, ΔAOB ~ ΔCOD. Hence ∠CAB = ∠ACD (alternate interior angles). So AB || CD. Therefore ABCD is a trapezium.

YEAR: 2023
CHAPTER: 7 - Triangles
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: State Basic Proportionality Theorem and use it to prove: A line through the mid-point of one side of a triangle, parallel to another side, bisects the third side.
ANSWER: BPT: If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. Given: In ΔABC, P is mid-point of AB and PQ || BC. By BPT: AP/PB = AQ/QC. Since AP = PB, AP/PB = 1 => AQ/QC = 1 => AQ = QC. Hence Q is mid-point of AC.

YEAR: 2023
CHAPTER: 7 - Triangles
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: State the converse of Basic Proportionality Theorem and use it to prove: Line segment joining mid-points of any two sides of a triangle is parallel to the third side.
ANSWER: Converse of BPT: If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side. Given: In ΔABC, D and E are mid-points of AB and AC. AD/DB = 1, AE/EC = 1 => AD/DB = AE/EC. By converse of BPT, DE || BC.

YEAR: 2023
CHAPTER: 7 - Triangles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In the given figure, ABC is a triangle in which AD = 1.6 cm, BD = 4.8 cm, AE = 1.1 cm and EC = 2.2 cm. Then:
OPTIONS: (A) DE || BC (B) DE = 1/2 BC (C) DE = BC (D) DE is not parallel to BC
ANSWER: (D) DE is not parallel to BC

YEAR: 2023
CHAPTER: 7 - Triangles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In two triangles ΔPQR and ΔABC, it is given that AB/BC = PQ/PR. For these two triangles to be similar, which of the following should be true?
OPTIONS: (A) ∠A = ∠P (B) ∠B = ∠Q (C) ∠B = ∠P (D) CA = QR
ANSWER: (C) ∠B = ∠P

YEAR: 2023
CHAPTER: 7 - Triangles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: ΔABC ~ ΔDEF and their perimeters are 32 cm and 24 cm respectively. If AB = 10 cm, then DE equals:
OPTIONS: (A) 8 cm (B) 7.5 cm (C) 15 cm (D) 5√3 cm
ANSWER: (B) 7.5 cm

Chapter 8: Circles

YEAR: 2023
CHAPTER: 8 - Circles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In the given figure, BC and BD are tangents to the circle with centre O and radius 9 cm. If OB = 15 cm, then the length (BC + BD) is:
OPTIONS: (A) 18 cm (B) 12 cm (C) 24 cm (D) 36 cm
ANSWER: (C) 24 cm

YEAR: 2023
CHAPTER: 8 - Circles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The distance between two parallel tangents of a circle of diameter 7 cm is:
OPTIONS: (A) 7 cm (B) 14 cm (C) 7/2 cm (D) 28 cm
ANSWER: (A) 7 cm

YEAR: 2023
CHAPTER: 8 - Circles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: From a point P, two tangents PQ and PR are drawn to a circle with centre at O. T is a point on the major arc QR of the circle. If $\angle QPR = 50°$, then $\angle QTR$ equals:
OPTIONS: (A) 50° (B) 130° (C) 65° (D) 90°
ANSWER: (C) 65°

YEAR: 2023
CHAPTER: 8 - Circles
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Prove that the tangents drawn from an external point to a circle are equal in length.

ANSWER: Given: Circle with centre O, external point P, tangents PQ and PR. Join OP, OQ, OR. In ΔOPQ and ΔOPR: OP = OP (common), OQ = OR (radii), ∠OQP = ∠ORP = 90°. By RHS congruence, ΔOPQ ≅ ΔOPR. Hence PQ = PR.

YEAR: 2023
CHAPTER: 8 - Circles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In the given figure, PQ and PR are tangents drawn from P to the circle with centre O such that ∠QPR = 65°. The measure of ∠QOR is:
OPTIONS: (A) 65° (B) 125° (C) 115° (D) 90°
ANSWER: (C) 115°

YEAR: 2023
CHAPTER: 8 - Circles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A circle of radius 5.2 cm has two tangents AB and CD parallel to each other. What is the distance between the two tangents?
OPTIONS: (A) 5.2 cm (B) 10.4 cm (C) 20.8 cm (D) can't find
ANSWER: (B) 10.4 cm

YEAR: 2023
CHAPTER: 8 - Circles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: In the given figure, O is the centre of the circle and PA is a tangent to the circle. If ∠OAB = 60°, then ∠OPA is equal to:
OPTIONS: (A) 60° (B) 30° (C) 15° (D) 20°
ANSWER: (B) 30°

Chapter 9: Introduction to Trigonometry

YEAR: 2023

CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: If 2 cos θ = 1, then the value of θ is
OPTIONS: (A) 45° (B) 60° (C) 30° (D) 90°
ANSWER: (B) 60°

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The value of 2 sin^2 30° + 3 tan^2 60° - cos^2 45° is:
OPTIONS: (A) 3√3 (B) 19/2 (C) 9/4 (D) 9
ANSWER: (D) 9

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: 9 sec^2 A - 9 tan^2 A is equal to:
OPTIONS: (A) 9 (B) 0 (C) 8 (D) 1/9
ANSWER: (A) 9

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: If tan A = 2/5, then the value of (1 - cos^2 A)/(1 - sin^2 A) is:
OPTIONS: (A) 25/4 (B) 4/25 (C) 4/5 (D) 5/4
ANSWER: (B) 4/25

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: 8(cos^2 A + sin^2 A) is equal to:

OPTIONS: (A) 1 (B) 0 (C) 9 (D) 8
ANSWER: (D) 8

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Evaluate: $2(\sin^2 45^\circ + \cot^2 30^\circ) - 6(\cos^2 45^\circ - \tan^2 30^\circ)$
ANSWER: $2[(1/2)+3] - 6[(1/2)-(1/3)] = 2(7/2) - 6(1/6) = 7 - 1 = 6$

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Evaluate: $\tan^2 60^\circ - 2 \operatorname{cosec}^2 30^\circ - 2 \tan^2 30^\circ$
ANSWER: $(\sqrt{3})^2 - 2(2)^2 - 2(1/\sqrt{3})^2 = 3 - 8 - 2/3 = -17/3$

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Evaluate: $(3/2)\tan^2 30^\circ - 2 \cos^2 90^\circ - (1/2)\operatorname{cosec}^2 30^\circ$
ANSWER: $(3/2)(1/3) - 0 - (1/2)(4) = 1/2 - 2 = -3/2$

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Prove that $\sec \theta (1 - \sin \theta)(\sec \theta + \tan \theta) = 1$
ANSWER: LHS = $(1/\cos \theta)(1 - \sin \theta)(1/\cos \theta + \sin \theta/\cos \theta) = (1/\cos \theta)(1 - \sin \theta)(1 + \sin \theta)/\cos \theta$ $= (1 - \sin^2 \theta)/\cos^2 \theta = \cos^2 \theta/\cos^2 \theta = 1$ = RHS.

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry

ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Prove that (1 + sec θ)/sec θ = sin^2 θ/(1 - cos θ)
ANSWER: LHS = (1 + 1/cos θ)/(1/cos θ) = 1 + cos θ = (1 - cos θ)(1 + cos θ)/(1 - cos θ) = (1 - cos^2 θ)/(1 - cos θ) = sin^2 θ/(1 - cos θ) = RHS.

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Prove that (sin A + cosec A)^2 + (cos A + sec A)^2 = 7 + tan^2 A + cot^2 A
ANSWER: LHS = sin^2 A + cosec^2 A + 2 + cos^2 A + sec^2 A + 2 = (sin^2 A + cos^2 A) + (1+cot^2 A) + (1+tan^2 A) + 4 = 1+1+cot^2 A+1+tan^2 A+4 = 7+tan^2 A+cot^2 A = RHS.

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Prove that cos A/(1+sin A) + (1+sin A)/cos A = 2 sec A
ANSWER: LHS = [cos^2 A + (1+sin A)^2]/[(1+sin A)cos A] = [cos^2 A + 1 + 2sin A + sin^2 A]/[(1+sin A)cos A] = [2 + 2sin A]/[(1+sin A)cos A] = 2/cos A = 2 sec A = RHS.

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: If sin α = 1/2, find the value of (3 cos α - 4 cos^3 α).
ANSWER: α = 30°. 3 cos 30° - 4 cos^3 30° = 3(√3/2) - 4(√3/2)^3 = 3√3/2 - 3√3/2 = 0.

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Prove that (1 + tan^2 A)/(1 + cot^2 A) = sec^2 A - 1

ANSWER: LHS = (1 + sin^2 A/cos^2 A)/(1 + cos^2 A/sin^2 A) = [(cos^2 A + sin^2 A)/cos^2 A]/[(sin^2 A + cos^2 A)/sin^2 A] = (1/cos^2 A)/(1/sin^2 A) = sin^2 A/cos^2 A = tan^2 A = sec^2 A - 1 = RHS.

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: Prove that (cot A - cos A)/(cot A + cos A) = cos^2 A/(1 + sin A)^2
ANSWER: LHS = [(cos A/sin A - cos A)/(cos A/sin A + cos A)] = [cos A(1/sin A - 1)]/[cos A(1/sin A + 1)] = (1 - sin A)/(1 + sin A) = (1 - sin A)(1 + sin A)/(1 + sin A)^2 = (1 - sin^2 A)/(1 + sin A)^2 = cos^2 A/(1 + sin A)^2 = RHS.

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Evaluate: 5 cosec^2 45° - 3 sin^2 90° + 5 cos 0°
ANSWER: 5(√2)^2 - 3(1)^2 + 5(1) = 10 - 3 + 5 = 12

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: Evaluate: 5 cosec^2 30° - cos 90° / 4 tan^2 60°
ANSWER: (5(2)^2 - 0)/(4(√3)^2) = 20/12 = 5/3

Chapter 10: Heights and Distances

YEAR: 2023
CHAPTER: 10 - Heights and Distances
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1

QUESTION: The angle subtended by a vertical pole of height 100 m at a point on the ground $100\sqrt{3}$ m from the base is, has measure of
OPTIONS: (A) 90° (B) 60° (C) 45° (D) 30°
ANSWER: (D) 30°

YEAR: 2023
CHAPTER: 10 - Heights and Distances
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The angle subtended by a tower of height 200 metres at a point 200 metres from the base is
OPTIONS: (A) 30° (B) 45° (C) 60° (D) 0°
ANSWER: (B) 45°

YEAR: 2023
CHAPTER: 10 - Heights and Distances
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The string of a kite in air is 50 m long and it makes an angle of 60° with the horizontal. Assuming the string to be straight, the height of the kite from the ground is:
OPTIONS: (A) $50\sqrt{3}$ m (B) $100/\sqrt{3}$ m (C) $50/\sqrt{3}$ m (D) $25\sqrt{3}$ m
ANSWER: (D) $25\sqrt{3}$ m

YEAR: 2023
CHAPTER: 10 - Heights and Distances
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: A TV tower stands vertically on the bank of a canal. From a point on the other bank directly opposite the tower, the angle of elevation of the top of the tower is 60°. From another point 20 m away from the point on the line joining this point to the foot of the tower, the angle of elevation of the top of the tower is 30°. Find the height of the tower.
ANSWER: Let height = h, width of canal = x. $\tan 60° = h/x \Rightarrow h = \sqrt{3}x$. $\tan 30° = h/(x+20) \Rightarrow 1/\sqrt{3} = h/(x+20) \Rightarrow x+20 = \sqrt{3}h = \sqrt{3}(\sqrt{3}x) = 3x \Rightarrow 2x = 20 \Rightarrow x = 10$. $h = 10\sqrt{3}$ m.

YEAR: 2023
CHAPTER: 10 - Heights and Distances

ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: An aeroplane when flying at a height of 4000 m from the ground passes vertically above another aeroplane at an instant when the angles of elevation of the two planes from the same point on the ground are 60° and 45° respectively. Find the vertical distance between the aeroplanes at that instant.
ANSWER: Let distance from point = x. For lower plane: $\tan 45^\circ = h/x \Rightarrow h = x$. For higher plane: $\tan 60^\circ = 4000/x \Rightarrow x = 4000/\sqrt{3}$. So $h = 4000/\sqrt{3}$. Distance = $4000 - 4000/\sqrt{3} = 4000(1 - 1/\sqrt{3}) = 1693.33$ m.

YEAR: 2023
CHAPTER: 10 - Heights and Distances
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: From the top of a building 60 m high, the angles of depression of the top and bottom of a tower are observed to be 30° and 60° respectively. Find the height of the tower. Also, find the distance between the building and the tower.
ANSWER: Let distance = x, height of tower = h. $\tan 60^\circ = 60/x \Rightarrow x = 60/\sqrt{3} = 20\sqrt{3}$. $\tan 30^\circ = (60-h)/x \Rightarrow 60-h = x/\sqrt{3} = 20 \Rightarrow h = 40$ m. Distance = 34.64 m.

YEAR: 2023
CHAPTER: 10 - Heights and Distances
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: The angle of elevation of the top of a building from a point A on the ground is 30°. On moving a distance of 30 m towards its base to the point B, the angle of elevation changes to 45°. Find the height of the building and the distance of its base from point A.
ANSWER: Let height = h, distance from B to building = x. $\tan 45^\circ = h/x \Rightarrow h = x$. $\tan 30^\circ = h/(x+30) \Rightarrow 1/\sqrt{3} = x/(x+30) \Rightarrow x+30 = \sqrt{3}x \Rightarrow x(\sqrt{3}-1) = 30 \Rightarrow x = 30/(\sqrt{3}-1) = 15(\sqrt{3}+1) = 40.98$ m. $h = 40.98$ m. Distance from A = $x+30 = 70.98$ m.

Chapter 11: Areas Related to Circles

YEAR: 2023
CHAPTER: 11 - Areas Related to Circles
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1
QUESTION: The area of a sector of angle α (in degrees) of a circle with radius R is:
OPTIONS: (A) α/180 × 2πR (B) α/360 × 2πR (C) α/180 × πR^2 (D) α/360 × πR^2
ANSWER: (D) α/360 × πR^2

YEAR: 2023
CHAPTER: 11 - Areas Related to Circles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The length of the arc of a circle of radius 14 cm which subtends an angle of 60° at the centre of the circle is:
OPTIONS: (A) 44/3 cm (B) 88/3 cm (C) 308/3 cm (D) 616/3 cm
ANSWER: (A) 44/3 cm

YEAR: 2023
CHAPTER: 11 - Areas Related to Circles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Area of a quadrant of a circle of radius 7 cm is
OPTIONS: (A) 154 cm^2 (B) 77 cm^2 (C) 77/2 cm^2 (D) 77/4 cm^2
ANSWER: (C) 77/2 cm^2

YEAR: 2023
CHAPTER: 11 - Areas Related to Circles
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: In the given figure, two concentric circles with centre O are shown. Radii of the circles are 2 cm and 5 cm respectively. Find the area of the shaded region.
ANSWER: Area of sector = $(60/360)\pi(5^2 - 2^2) = (1/6)\pi(25-4) = (1/6)\times(22/7)\times21 = 11$ cm^2.

YEAR: 2023
CHAPTER: 11 - Areas Related to Circles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The area of a sector of angle α (in degrees) of a circle with radius R is:
OPTIONS: (A) α/180 × 2πR (B) α/360 × 2πR (C) α/180 × πR^2 (D) α/360 × πR^2

ANSWER: (D) $\alpha/360 \times \pi R^2$

YEAR: 2023
CHAPTER: 11 - Areas Related to Circles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The length of arc of a circle of radius 7 cm which subtends an angle of 90° at the centre of the circle is:
OPTIONS: (A) 22 cm (B) 11 cm (C) 77/2 cm (D) 11/2 cm
ANSWER: (B) 11 cm

Chapter 12: Surface Areas and Volumes

YEAR: 2023
CHAPTER: 12 - Surface Areas and Volumes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The volume of a cone of radius r and height 3r is:
OPTIONS: (A) $1/3\ \pi r^3$ (B) $3\pi r^3$ (C) $9\pi r^3$ (D) πr^3
ANSWER: (D) πr^3

YEAR: 2023
CHAPTER: 12 - Surface Areas and Volumes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A solid is of the form of a cone of radius r surmounted on a hemisphere of the same radius. If the height of the cone is the same as the diameter of its base, then the volume of the solid is:
OPTIONS: (A) πr^3 (B) $4/3\ \pi r^3$ (C) $3\pi r^3$ (D) $2/3\ \pi r^3$
ANSWER: (B) $4/3\ \pi r^3$

YEAR: 2023
CHAPTER: 12 - Surface Areas and Volumes
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1
QUESTION: The curved surface area of a cone of radius 7 cm is 550 cm^2. Its slant height is:
OPTIONS: (A) 24 cm (B) 25 cm (C) 14 cm (D) 20 cm
ANSWER: (B) 25 cm

YEAR: 2023
CHAPTER: 12 - Surface Areas and Volumes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The curved surface area of a right circular cylinder of height 14 cm is 88 cm^2. The diameter of its circular base is:
OPTIONS: (A) 2 cm (B) 1 cm (C) 4 cm (D) 7 cm
ANSWER: (A) 2 cm

YEAR: 2023
CHAPTER: 12 - Surface Areas and Volumes
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: A vessel is in the form of a hemispherical bowl surmounted by a hollow cylinder of same diameter. The diameter of the hemispherical bowl is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area of the vessel. Also, find the volume of the vessel.
ANSWER: Radius = 7 cm, height of cylinder = 13-7 = 6 cm. Inner surface area = $2\pi rh + 2\pi r^2$ = $2\pi(7)(6) + 2\pi(49) = 84\pi + 98\pi = 182\pi$ = 572 cm^2. Volume = $\pi r^2 h + 2/3\pi r^3 = \pi(49)(6) + 2/3\pi(343) = 294\pi + 686\pi/3 = (882+686)\pi/3 = 1568\pi/3$ = 1642.67 cm^3.

YEAR: 2023
CHAPTER: 12 - Surface Areas and Volumes
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: The sum of the radius of the base and height of a solid right-circular cylinder is 37 cm. If the total surface area of the solid cylinder is 1628 cm^2, find the volume of the cylinder.
ANSWER: r + h = 37. Total surface area = $2\pi r(r+h) = 1628 \Rightarrow 2\pi r(37) = 1628 \Rightarrow r = 1628/(2\times22/7\times37) = 1628\times7/(44\times37) = 7$ cm. h = 30 cm. Volume = $\pi r^2 h = (22/7)\times49\times30 = 4620$ cm^3.

YEAR: 2023

CHAPTER: 12 - Surface Areas and Volumes
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: A heap of rice is in the form of a cone of base diameter 24 m and height 7/2 m. Find the volume of rice. How much canvas cloth is required to just cover the heap?
ANSWER: Radius = 12 m, h = 7/2 m. Volume = $1/3\pi r^2 h$ = 1/3 × 22/7 × 144 × 7/2 = 528 m^3. Slant height = $\sqrt{(h^2+r^2)}$ = $\sqrt{(49/4+144)}$ = $\sqrt{(625/4)}$ = 25/2 m. Canvas required = πrl = 22/7 × 12 × 25/2 = 3300/7 = 471.43 m^2.

YEAR: 2023
CHAPTER: 12 - Surface Areas and Volumes
ORIGINAL_SECTION: E
ORIGINAL_MARKS: 4
QUESTION: A wooden toy is shown in the picture. This is a cuboidal wooden block of dimensions 14 cm × 17 cm × 4 cm. On its top there are seven cylindrical hollows for bees to fit in. Each cylindrical hollow is of height 3 cm and radius 2 cm. (i) Find the volume of wood carved out to make one cylindrical hollow. (ii) Find the lateral surface area of the cuboid to paint it with green colour. (iii)(a) Find the volume of wood in the remaining cuboid after carving out seven cylindrical hollows. OR (iii)(b) Find the surface area of the top surface of the cuboid to be painted yellow.
ANSWER: (i) Volume of one hollow = $\pi r^2 h$ = 22/7 × 4 × 3 = 264/7 = 37.7 cm^3. (ii) LSA = 2h(l+b) = 2×4(14+17) = 248 cm^2. (iii)(a) Volume of 7 hollows = 7×264/7 = 264 cm^3. Volume of cuboid = 14×17×4 = 952 cm^3. Remaining = 952 - 264 = 688 cm^3. (b) Top surface area = l×b - $7\pi r^2$ = 238 - 7×22/7×4 = 238 - 88 = 150 cm^2.

YEAR: 2023
CHAPTER: 12 - Surface Areas and Volumes
ORIGINAL_SECTION: E
ORIGINAL_MARKS: 4
QUESTION: Flower beds look beautiful growing in gardens. One such circular park of radius r m, has two segments with flowers. One segment which subtends an angle of 90° at the centre is full of red roses, while the other segment with central angle 60° is full of yellow coloured flowers. The combined area is 256⅔ sq m. (i) Write an equation representing the total area of the two segments in terms of r. (ii) Find the value of r. (iii)(a) Find the area of the segment with red roses. OR (iii)(b) Find the area of the segment with yellow flowers.
ANSWER: (i) Area = $(90/360)\pi r^2 - (1/2)r^2 + (60/360)\pi r^2 - (\sqrt{3}/4)r^2$ = 256⅔. (ii) r ≈ 26.1 cm. (iii)(a) Area of red segment = $(1/4)\pi r^2 - (1/2)r^2$ ≈ 194.63 cm^2. (b) Area of yellow segment = $(1/6)\pi r^2 - (\sqrt{3}/4)r^2$ ≈ 62.03 cm^2.

YEAR: 2023
CHAPTER: 12 - Surface Areas and Volumes
ORIGINAL_SECTION: E
ORIGINAL_MARKS: 4
QUESTION: Interschool Rangoli Competition: Rangoli is in the shape of square marked as ABCD, side of square being 40 cm. At each corner of a square, a quadrant of circle of radius 10 cm is drawn. Also a circle of diameter 20 cm is drawn inside the square. (i) What is the area of square ABCD? (ii) Find the area of the circle. (iii) If the circle and the four quadrants are cut off from the square ABCD and removed, then find the area of remaining portion. OR (iii) Find the combined area of 4 quadrants and the circle removed.
ANSWER: (i) Area of square = 1600 cm^2. (ii) Area of circle = $\pi(10)^2$ = 2200/7 = 314.28 cm^2. (iii) Area of 4 quadrants = $4 \times 1/4 \times \pi \times 100$ = 2200/7 = 314.28 cm^2. Remaining area = 1600 - 4400/7 = 6800/7 = 971.43 cm^2. Combined area = 4400/7 = 628.57 cm^2.

Chapter 13: Statistics

YEAR: 2023
CHAPTER: 13 - Statistics
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Median and Mode of a distribution are 25 and 21 respectively. Mean of the data using empirical relationship is:
OPTIONS: (A) 27 (B) 29 (C) 18 (D) 29/3
ANSWER: (A) 27

YEAR: 2023
CHAPTER: 13 - Statistics
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The graph of y = p(x) is shown in the figure for some polynomial p(x). The number of zeroes of p(x) is/are:
OPTIONS: (A) 0 (B) 1 (C) 2 (D) 3
ANSWER: (A) 0

YEAR: 2023

CHAPTER: 13 - Statistics
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: The table given below shows the daily expenditure on food of 25 households in a locality. Find the mean daily expenditure on food. Also, find the mode of the data.
Daily expenditure (₹): 100-150, 150-200, 200-250, 250-300, 300-350
Number of households: 4, 5, 12, 2, 2
ANSWER: Mean = 5275/25 = 211. Modal class = 200-250. Mode = 200 + (12-5)/(24-5-2) × 50 = 200 + 7/17 × 50 = 200 + 350/17 = 220.59.

YEAR: 2023
CHAPTER: 13 - Statistics
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: The distribution below gives the weights of 30 students of a class. Find the median weight of the students.
Weight: 40-45, 45-50, 50-55, 55-60, 60-65, 65-70, 70-75
Number: 2, 3, 8, 6, 6, 3, 2
ANSWER: N = 30, N/2 = 15. Median class = 55-60. Median = 55 + (15-13)/6 × 5 = 55 + 10/6 = 56.67.

YEAR: 2023
CHAPTER: 13 - Statistics
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The mode of the numbers 2, 3, 3, 4, 5, 4, 4, 5, 3, 4, 2, 6, 7 is:
OPTIONS: (A) 2 (B) 3 (C) 4 (D) 5
ANSWER: (C) 4

YEAR: 2023
CHAPTER: 13 - Statistics
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The median of first seven prime numbers is:
OPTIONS: (A) 5 (B) 7 (C) 11 (D) 13
ANSWER: (B) 7

YEAR: 2023
CHAPTER: 13 - Statistics
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The mean of first ten natural numbers is:
OPTIONS: (A) 5.5 (B) 55 (C) 45 (D) 4.5
ANSWER: (A) 5.5

YEAR: 2023
CHAPTER: 13 - Statistics
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: If the mean of 6, 7, x, 8, y, 14 is 9, then:
OPTIONS: (A) x+y=21 (B) x+y=19 (C) x-y=19 (D) x-y=21
ANSWER: (B) x+y=19

YEAR: 2023
CHAPTER: 13 - Statistics
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3
QUESTION: A survey conducted on 20 households in a locality resulted in the following frequency table for the number of family members. Find the median of this data.
Family size: 1-3, 3-5, 5-7, 7-9, 9-11
Number of families: 7, 8, 2, 2, 1
ANSWER: N=20, N/2=10. Median class = 3-5. Median = 3 + (10-7)/8 × 2 = 3 + 3/4 = 3.75.

YEAR: 2023
CHAPTER: 13 - Statistics
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: Find the mean and the median of the following data:
Class: 0-10, 10-20, 20-30, 30-40, 40-50, 50-60, 60-70, 70-80
Frequency: 3, 5, 16, 12, 13, 20, 6, 5
ANSWER: Mean = 42. Median class = 40-50. Median = 40 + (40-36)/13 × 10 = 43.1.

YEAR: 2023
CHAPTER: 13 - Statistics
ORIGINAL_SECTION: D
ORIGINAL_MARKS: 5
QUESTION: Find the mean and the median of the following data:
Class: 85-90, 90-95, 95-100, 100-105, 105-110, 110-115
Frequency: 10, 12, 15, 14, 12, 7
ANSWER: Mean = 99.4. Median class = 95-100. Median = $95 + (35-22)/15 \times 5 = 99.33$.

YEAR: 2023
CHAPTER: 13 - Statistics
ORIGINAL_SECTION: E
ORIGINAL_MARKS: 4
QUESTION: Blood group: In a sample of 50 people, 21 had type O, 22 had type A, 5 had type B and rest had type AB. (i) What is the probability that a person chosen at random had type O blood? (ii) What is the probability that a person chosen at random had type AB blood group? (iii) What is the probability that a person chosen at random had neither type A nor type B blood group? OR (iii) What is the probability that person chosen at random had either type A or type B or type O blood group?
ANSWER: (i) P(O) = 21/50. (ii) Type AB = 50-21-22-5 = 2. P(AB) = 2/50 = 1/25. (iii) P(neither A nor B) = (21+2)/50 = 23/50. OR P(A or B or O) = (21+22+5)/50 = 48/50 = 24/25.

Chapter 14: Probability

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A card is drawn at random from a well-shuffled deck of 52 cards. The probability of getting a red card is:
OPTIONS: (A) 1/26 (B) 1/13 (C) 1/4 (D) 1/2
ANSWER: (D) 1/2

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1
QUESTION: A die is thrown once. Find the probability of getting a number less than 7.
OPTIONS: (A) 5/6 (B) 1 (C) 1/6 (D) 0
ANSWER: (B) 1

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A card is drawn at random from a well-shuffled deck of 52 playing cards. The probability of getting an ace of spade is:
OPTIONS: (A) 1/13 (B) 3/52 (C) 1/26 (D) 1/52
ANSWER: (D) 1/52

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A die is thrown once. The probability of getting an odd prime number is:
OPTIONS: (A) 1/2 (B) 1/6 (C) 1/3 (D) 2/3
ANSWER: (C) 1/3

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: From a well-shuffled deck of 52 playing cards, all diamond cards are removed. Now, a card is drawn from the remaining pack at random. Find the probability that the selected card is a king.
ANSWER: Total cards = 52-13 = 39. Kings remaining = 3. P(king) = 3/39 = 1/13.

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: C
ORIGINAL_MARKS: 3

QUESTION: A die is rolled once. Find the probability of getting: (i) an even prime number (ii) a number greater than 4 (iii) an odd number.
ANSWER: S = {1,2,3,4,5,6}. (i) Even prime = 2 => P = 1/6. (ii) Numbers > 4 = 5,6 => P = 2/6 = 1/3. (iii) Odd numbers = 1,3,5 => P = 3/6 = 1/2.

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Two coins are tossed together. The probability of getting atmost two heads, is:
OPTIONS: (A) 1/2 (B) 1/4 (C) 3/4 (D) 1
ANSWER: (D) 1

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Which of the following numbers cannot be the probability of an event?
OPTIONS: (A) 0.5 (B) 5% (C) 1/0.5 (D) 0.5/14
ANSWER: (C) 1/0.5

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A box contains 90 discs, numbered from 1 to 90. If one disc is drawn at random, then find the probability that it bears a multiple of 15.
ANSWER: Multiples of 15: 15,30,45,60,75,90 = 6. P = 6/90 = 1/15.

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: A card is drawn at random from a well-shuffled deck of 52 playing cards. What is the probability of getting a black king?
OPTIONS: (A) 1/26 (B) 1/13 (C) 1/52 (D) 1/2

ANSWER: (A) 1/26

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: B
ORIGINAL_MARKS: 2
QUESTION: A bag contains 30 discs numbered from 1 to 30. One disc is drawn at random from the bag. Find the probability that it bears a number (a) divisible by 6 (b) greater than 25.
ANSWER: (a) Numbers divisible by 6: 6,12,18,24,30 = 5. P = 5/30 = 1/6. (b) Numbers > 25: 26,27,28,29,30 = 5. P = 5/30 = 1/6.

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Two dice are rolled together. The probability that the sum of the numbers that appeared is 9, is:
OPTIONS: (A) 5/36 (B) 1/9 (C) 1/12 (D) 1/6
ANSWER: (B) 1/9

Assertion-Reason Questions (Chapter-wise distribution)

YEAR: 2023
CHAPTER: 8 - Circles
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): A tangent to a circle is perpendicular to the radius through the point of contact. Reason (R): The lengths of tangents drawn from the external point to a circle are equal.
OPTIONS: (A) Both true and R is correct explanation (B) Both true but R not correct explanation (C) A true, R false (D) A false, R true
ANSWER: (B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A)

YEAR: 2023
CHAPTER: 3 - Pair of Linear Equations in Two Variables
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): The system of linear equations $3x + 5y - 4 = 0$ and $15x + 25y - 25 = 0$ is inconsistent. Reason (R): The pair of linear equations $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$ is inconsistent if $a_1/a_2 = b_1/b_2 \neq c_1/c_2$.
OPTIONS: (A) Both true and R is correct explanation (B) Both true but R not correct explanation (C) A true, R false (D) A false, R true
ANSWER: (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

YEAR: 2023
CHAPTER: 2 - Polynomials
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): Polynomial $x^2 + 4x$ has two real zeroes. Reason (R): Zeroes of the polynomial $x^2 + ax$ $(a\neq0)$ are 0 and a.
OPTIONS: (A) Both true and R is correct explanation (B) Both true but R not correct explanation (C) A true, R false (D) A false, R true
ANSWER: (C) Assertion (A) is true, but Reason (R) is false.

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): The probability of getting a prime number, when a die is thrown once, is 2/3. Reason (R): On the faces of a die, prime numbers are 2, 3, 5.
OPTIONS: (A) Both true and R is correct explanation (B) Both true but R not correct explanation (C) A true, R false (D) A false, R true
ANSWER: (D) Assertion (A) is false, but Reason (R) is true.

YEAR: 2023
CHAPTER: 4 - Quadratic Equations
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1

QUESTION: Assertion (A): If one root of the quadratic equation $4x^2 - 10x + (k-4) = 0$ is reciprocal of the other, then value of k is 8. Reason (R): Roots of the quadratic equation $x^2 - x + 1 = 0$ are real.
OPTIONS: (A) Both true and R is correct explanation (B) Both true but R not correct explanation (C) A true, R false (D) A false, R true
ANSWER: (C) Assertion (A) is true, but Reason (R) is false.

YEAR: 2023
CHAPTER: 12 - Surface Areas and Volumes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): The surface area of largest sphere that can be inscribed in a hollow cube of side a cm is πa^2 cm^2. Reason (R): The surface area of a sphere of radius r is $4/3\pi r^3$.
OPTIONS: (A) Both true and R is correct explanation (B) Both true but R not correct explanation (C) A true, R false (D) A false, R true
ANSWER: (C) Assertion (A) is true, but Reason (R) is false.

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): When two coins are tossed together, the probability of getting no tail is 1/4. Reason (R): The probability P(E) of an event E satisfies $0 \leq P(E) \leq 1$.
OPTIONS: (A) Both true and R is correct explanation (B) Both true but R not correct explanation (C) A true, R false (D) A false, R true
ANSWER: (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

YEAR: 2023
CHAPTER: 9 - Introduction to Trigonometry
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): For $0 < \theta \leq 90°$, cosec θ - cot θ and cosec θ + cot θ are reciprocal of each other. Reason (R): $\cot^2\theta - \mathrm{cosec}^2\theta = 1$.
OPTIONS: (A) Both true and R is correct explanation (B) Both true but R not correct explanation (C) A true, R false (D) A false, R true
ANSWER: (C) Assertion (A) is true, but Reason (R) is false.

YEAR: 2023
CHAPTER: 14 - Probability
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Assertion (A): The probability that a leap year has 53 Sundays is 2/7. Reason (R): The probability that a non-leap year has 53 Sundays is 1/7.
OPTIONS: (A) Both true and R is correct explanation (B) Both true but R not correct explanation (C) A true, R false (D) A false, R true
ANSWER: (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
